

Jacob Groner – AI Research Engineer

jacobgroner@gmail.com • jacobgroner.com

Education

Cornell University, College of Arts & Sciences, Ithaca, NY

Graduated December 2025

BA in Computer Science and Mathematics, Minor in Biological Sciences

Machine Learning • AI Reasoning/Decision-Making • Reinforcement Learning • Program Synthesis • Discrete Structures • Analysis of Algorithms • Linear Algebra • Multivariable Calculus • Statistics • Differential Equations

Selected Projects

[MIRAGE](#) – LLM enhancement to improve generalization

- Designed dual-process architecture combining a meta-trained transformer decomposer with a schema-based reasoning engine for compositional sequence modeling
- Achieved 99.6% accuracy on SCAN by enabling iterative decomposition and explicit schema application, supporting systematic out-of-distribution generalization

[Quoridor Bot](#) – Search-Based Game Agent

- Implemented Monte Carlo Tree Search agent for adversarial board game, modeling implementation after AlphaGo
- Integrated convolutional neural network evaluation function for early-cutoff

Experience

NeuroAI Research, Lab of Professor Weinan Sun, Ithaca, NY

May 2024 – February 2026

- Implemented wavefront planning algorithm over learned latent state space of a cloned Hidden Markov model (CSCG), enabling goal-directed navigation in partially observed environments
- Analyzed hippocampal calcium imaging data using spatial binning, UMAP, and spectral methods (Fourier, Lomb–Scargle) to identify place cells and visualize the emergence of neural representations during reward-guided learning
- Built Unity pipeline to generate VR mazes for head-fixed mouse navigation experiments, included MQTT communication protocol which allowed for behavioral data collection during optical imaging

Neuroscience Research, Lab of Professor Ian Ellwood, Ithaca, NY

August 2021 – November 2022

- Led design of whisker-based navigation system for head-fixed mice, implementing real-time control algorithm to map virtual maze position to motor-driven tactile feedback via dynamic wheel actuation

Software Developer, Firefly Neuroscience, Tel Aviv, Israel

May 2022 – August 2022

- Developed LSTM-based sequence model for detecting muscle and eye artifacts in long-horizon multichannel EEG recordings
- Built EEG augmentation pipeline using random time-window recombination and smooth interpolation to reduce artifact-model overfitting

Instructor, ID Tech, Irvine, CA

June 2021 – August 2021

- Taught week-long Java and Python courses to cohorts of 8 students, guiding them from core programming concepts to building and presenting independent projects

Skills

Workflow Preferences: OpenAI Gym • PyTorch • Jax • Pandas • Numpy • Plotly • VSCode • Windows • Google Drive • Codex

ML Techniques: DQN • PPO • LSTM • Transformers • RAG • CNN • VAE • HMM • EM • A* • MCTS • PCA

Languages: English (native) • Hebrew (Conversational)

Hobbies

Surfing • Chess • Ultimate Frisbee • Bass Guitar